

The Bridge Certificate Applied to the Joshi ATS Series: A Ledger Audit

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2 October 2026 — Version 0.3 (working paper)

Abstract

We revisit the pre-specified form audit of Kirti Joshi’s Arithmetic Teichmüller Space (ATS) series, with source versions and the historical selection of rows R2, R4, R6, R8 preserved. This is a selected-rows form audit, not a complete certificate or a proof-correctness audit. The source makes the j^2 -scaling law, the assigned compensating site, and the asymmetric Frobenius shift visible. These historical findings have narrower predicates than the current seven-component contract: locating a load is not a quantified budget, and seeing a second mode is not a complete E8 mode audit. The revised crosswalk assigns F-R2a to C4 (dependent on C6), F-R2b and F-R6a to C6, and F-R6b to C7; F-R8 belongs to the triggered extension E8 outside that crosswalk. A declared normalization convention is classified as partially delivering (I), rather than as a derivation of cross-container compatibility. We correct our previous generic assertion that a set’s volume dominates the absolute value of each element: the cited ATS III argument concerns a tensor-product box under specific integral and weight hypotheses. A simple translated-ball example refutes only our generic paraphrase. ATS III box containment and ATS IV’s normalization and measure bridge remain separate obligations. Later normalization and effective-weight classifications are cited within their stated hypotheses; they do not close indexed log-volume, hull, measure, or mode compatibility. No claim about the truth of abc, the correctness of IUT, or the correctness of the ATS series follows from this audit.

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1 Mandate, Source Versions, and Scope

This paper is Part 4 of the abc Gap Series. Part 1 [8] analyzed the comparison in the transport step of Corollary 3.12 of IUTchIII [6]. Part 2 [9] defines the *bridge contract*; Part 3 [10] supplies its line-level ledger instrument. Both are published working papers. The present paper applies selected historical predicates to the exact ATS versions [1, 2, 3, 4, 5]. The correction dated 2 October 2026 processes the inventoried mapping, source-paraphrase, and status-import errors in our v0.2 text; it performs no new whole-series proof audit.

1.1 Ground rules

First, **no authority argument**: neither “Joshi asserts it holds” nor “Scholze–Stix [7] assert it does not” counts as evidence; every verdict is anchored in the verbatim wording of the audited sources. Joshi’s own status tables about the Scholze–Stix objections are treated as self-report, not as evidence. Second, **no truth verdict**: the audit judges whether the text serves pre-specified contract clauses, not whether abc is true, whether Joshi’s proof is correct, or whether the critics are right. Third, **status discipline**: the audited texts are publicly posted, versioned arXiv preprints, not peer-reviewed publications; ATS IV carries the header “Preliminary version for comments” and opens with “[This is still a work in progress!]”. We therefore write “publicly posted text” throughout, bind exact versions (Table 2), and treat the preprint status as a version-stability caveat recorded per finding.

1.2 Pre-specified scope: a four-row test

The audit mandate — fixed in the series’ ledger instrument before this test was run — restricts the test to four ledger rows. **This is a commissioned four-row test, not a complete F-R0–F-R7 core audit with triggered E8.** Rows R0, R1, R3, R5, R7 were not part of the mandate (R5 is touched only insofar as R4 forces it). Table 1 fixes the included and excluded rows.

Row	Reason for pre-selection	Not covered
R2	numerical pilot transport; comparative discrimination case	no R0/R1 audit
R4	visible treatment of the j^2 -load	R5 only via R4
R6	contract components 6–7; the central bridge question	no proof-correctness audit
F-R8	visibility of the second mode	no completeness of all auxiliary modes

Table 1: Pre-specified four-row audit scope. The table documents the mandate boundary and prevents the row findings from being read as a complete certificate audit.

1.3 Operational verdict rules

Each atomic claim receives one of three verdicts. *Delivers* = the pre-stated minimal predicate of the row is served by the quoted text. *Partially delivers* = at least one explicit sub-predicate remains unserved. *Does not deliver* = the minimal predicate is unserved. **A row-level pass is not a certificate pass**: it certifies only the row’s minimal predicate, never the full seven-component contract. Confidence is reported in three separate dimensions: **Q** (source binding: is the verbatim anchor unambiguous?), **K** (criterion fit: does the wording serve the pre-stated predicate?), **S** (substance status: checked / not checked / externally marked open).

Operational glossary (minimal pairs used below): *transported* = a numerical quantity together with an explicit transformation law; *invariant* = same output value after typed identification; *addressed* = load and its compensating site named; *quantitatively booked* = amount/rule balanced in a complete budget; *visible mode* = mode, role, and affected bound expressly named.

1.4 Mapping to the current contract

The historical mandate remains R2/R4/R6 plus R8. In the current namespace of Part 2 [9], the atomic findings below use F-R2a, F-R2b, F-R4a, F-R4b, F-R6a, F-R6b, and F-R8. F-R2a belongs to C4 with a dependency on C6; F-R2b and F-R6a to C6; F-R4a and F-R4b to C5; F-R6b to C7. The typed evaluation identity $\nu_C \circ i_q = \nu_q$ is a C6/F-R6a obligation, not component 1 and not certified merely by displaying a scaling law. F-R8 is the triggered E8 extension, outside the seven-component crosswalk. It is X (not included in a denominator) when no relevant multi-mode, sign, or ordering trigger occurs; the explicit Frobenius shift triggers it here.

The historical address/visibility predicates are retained as such. A historical *delivers* finding for F-R4a or F-R6b does not assert the current C5 budget or the current C7 hypothesis-by-hypothesis coverage. Likewise, F-R8 visibility does not establish complete E8 mode coverage. We therefore report no current full-certificate score and no ratio of successful rows. Under the current grammar, a documented convention is at most I (partially delivers), not E+ (delivers a compatibility equation). Joshi is a comparative discrimination case; it is not the accepted control of the separate historical v1-T illustration in Part 2.

1.5 Source manifest

1.6 Four “tautology” passages, disambiguated

The audited texts use the word “tautology” in four places with distinct scopes, and conflating them produces the false impression that Joshi calls his own

ID	Source (exact version audited)	Role
ATS III	arXiv:2401.13508v4 (rev. 24 Feb 2025) [4]	Cor. 3.12 construction (R2, R4, R6)
ATS IV	arXiv:2403.10430v2 (rev. 24 Feb 2025) [5]	abc conclusion, bridge site (R2, R6, F-R8)
ATS II($\frac{1}{2}$)	arXiv:2305.10398v12 [3]	normalization data (status notice A1)

Table 2: Audited sources. Line anchors “extract ll.” refer to the plain-text extracts of these exact PDF versions retained with the historical audit records. They are local finding aids, not independently published evidence; the exact-version section/theorem citations are the public primary-source anchors. Audit date: 12 August 2026; status notices dated 13/15 August 2026 (Section 3).

arithmetic proof a tautology — which he does not. (T-a) a measure-theoretic triviality used as a tool (ATS III, Lemma 9.10.2.2(4)); (T-b) a statement about the *geometric* analog only (ATS III, § 12.18); (T-c) a statement about Joshi’s own variant $\tilde{\Theta}_{\text{Joshi}}$, while the theorem in question is proven for $\tilde{\Theta}_{\text{Mochizuki}}$ (ATS IV, § 1.6(4e)); and (T-d) the load-bearing bridge equality in the proof of ATS IV, Theorem 6.10.1. **Only T-d is load-bearing for this audit.**

2 The Ledger Audit (Atomic Findings)

2.1 F-R2a — intra-link transport: delivers

Minimal predicate: the numerical q -readout is transported along the theta-link by an explicit numerical law, with the j^2 -factor carried visibly. ATS III, Theorem 4.2.2.1(4) states the valuation scaling relation $|-|_{K'_{y_w,j}} = |-|_{K'_{y_w,1}}^{j^2}$ at the odd semistable places (“Then one has the following relationship between their valuations,” eq. (4.2.2.2); extract ll. 1544–1560); § 9.9 carries the q -value through it explicitly (eqs. (9.9.3)–(9.9.4); extract ll. 6458–6484); and § 9.10.2 supplies the concrete log-volume scale (normalized by $\text{LogVol}(O_{L'_w}) = 0$, $\text{Vol}(\lambda O) = |\lambda|$; extract ll. 6562–6596). This is *transport by an explicit numerical law*, not invariance of a value: what is certified is a covariant transformation rule with a concrete numerical target scale, serving the historical F-R2a/C4 transport-law predicate (dependent on C6). The full typed identity $\nu_C \circ i_q = \nu_q$ is not written out in the source and is not certified here. **Verdict: delivers (row predicate). Q high / K high / S not checked.**

2.2 F-R2b — cross-container transport: set by convention

Minimal predicate: the equality of readouts across the global Frobenius shift $y_0 \leftrightarrow \varphi(y_0)$ is derived, not set. Here the equality is *set* by the choice of “the same global normalization” (ATS IV, Thm. 6.10.1 preamble; extract ll. 3186–3189), with a one-sentence proof (“The equality . . . is a consequence of the choice of the same global normalization . . .”; extract ll. 3262–3266), while the local constituents change under φ and the source itself notes that this “precludes direct comparison of various local quantities” (ATS IV, Rem. 6.10.2; extract ll. 3208–3211). Not a contradiction — but a convention, not a derived compatibility. **Verdict: partially delivers (I): a convention is documented; derived compatibility is unserved. Q high / K high / S see Section 3.** This is the current F-R2b/C6 classification, not an E+ compatibility finding.

2.3 F-R4a — the j^2 -load: addressed and assigned

Minimal predicate: the load is addressed — its origin, propagation, and compensating site are named; it does not vanish unbooked in one coordinate and return silently. The load arises only at the odd semistable places (Thm. 4.2.2.1, eqs. (4.2.2.2) vs. (4.2.2.3); extract ll. 1544–1570); it survives into the numerical evaluation (the $\sum_j j^2/\ell^{*2}$ of § 9.9; extract ll. 6476–6498); and its return is assigned to a forced renormalization at the complementary places, with the product formula as the booking instrument (Cor. 4.2.2.4, extract ll. 1600–1614; Thm. 4.6.1(4), extract ll. 1786–1791). The feared scenario — silent vanishing and unbooked return — does not occur in the text. **Verdict: delivers (row predicate). Q high / K high / S not checked.** The historical address predicate does not quantify all loads, signs, residuals, and compensating budget entries required by current C5; no current C5/E+ finding follows. A recorded citation inconsistency: the source refers to the same relation alternately as Theorem 4.2.2.1(3) and (4); the referent is unambiguous.

2.4 F-R4b — quantitative balance: open

Minimal predicate: the compensation is quantitatively booked — amount, uniqueness, and budget split shown. The existence of the compensating renormalization is inferred from the product-formula constraint in a one-line proof (“This is clear from Theorem 4.2.2.1 and the global constraint placed by the requirement that the product formula holds”; extract ll. 1613–1614). The residual load is *located but not quantified*; contract component 5 (separated indeterminacy budget) is served qualitatively, not quantitatively. **Verdict: partially delivers. Q high / K high / S not checked.**

2.5 F-R6b — order relation, form level: written-out proof with an addressable core step

Minimal predicate: the order relation is stated as a theorem with a written-out proof whose steps are addressable. ATS III, Theorem 9.11.1 (extract ll. 6794–6806; in Mochizuki’s notation Cor. 9.11.1.1, extract ll. 6882–6888) states the volume lower bound; the evaluation exists in the container as an adelic log-volume computed as the sum of local contributions (“one can compute $\log \text{Vol}(\dots)$ by taking the sum ... of the local contributions at all primes”; ATS IV, § 1.5, extract ll. 764–769); this is the contract component 6 evaluation audited in Section 2.6, not part of the order-relation finding recorded here. The cited mechanism is source-specific: ATS III, Lemma 9.10.7.1 (PDF p. 125) treats a superset S' of a tensor-product box, with integral $\tau_i \in O_{E_i}$, constituents $s_i \in \tau_i O_{E_i}$, and weights $0 < \gamma_i \leq 1$. Theorem 9.11.1 (PDF p. 127) uses the asserted inclusion of that box in the construction. Neither statement is the generic proposition that the volume of an arbitrary set bounds the absolute value of each element. Our v0.2 paraphrase of that kind was false: with normalized additive Haar measure $\mu(\mathbb{Z}_p) = 1$, the translated ball

$$U = 1 + p\mathbb{Z}_p, \quad 1 \in U, \quad \mu(U) = p^{-1} < |1|_p = 1$$

is a counterexample. It refutes our unrestricted paraphrase, not Joshi’s tensor-box lemma, Theorem 9.11.1, IUT, or abc. In particular, the centered-module hypothesis in ATS III, Lemma 9.10.2.2 (PDF p. 123) cannot be silently dropped.

The historical form finding is limited to a stated theorem and an addressable proof. Establishing that the actual constructed theta-set supplies the box, integral structure, and applicable weight hypotheses remains the separate A2 obligation. Named proof addresses alone do not discharge current C7: each load-bearing hypothesis, including the inclusion hypothesis, needs its own applicable anchor. The availability of adelic evaluation alone does not establish C6 compatibility. **Verdict: delivers the form predicate. Q high / K high / S externally marked open (A2).**

2.6 F-R6a — cross-container compatibility: set, and global-only by the source’s own restriction

Minimal predicate (contract component 6, global form): the connecting compatibility between the two evaluation routes is derived; contract component 7 in turn uses it to conclude $\nu_q(\text{Obj}_q) \in H_\Theta$ under admissible linkage. For the abc conclusion the lower bound is computed with $\varphi(y_0)$, the upper with y_0 ; they are connected by the middle equality of ATS IV, Theorem 6.10.1, whose complete proof reads: “The middle equality is a tautology using the fact that the number fields provided by y'_0 and y_0 are identically normalized” (T-d; extract ll. 3272–3273). Two clarifications keep the standard honest.

First, the certificate’s order clause is *global*; the source’s own restriction — one “cannot claim that $z_p \leq z'_p$ holds for all primes p without violating the middle equality” (ATS IV, § 1.5, extract ll. 816–819), and “one cannot make local term by term comparisons at each prime” (extract ll. 772–775) — is a *scope limit of the claim*, not a violated criterion. Second, what remains unserved at the form level is the *derivation* of the cross-container compatibility: it is set by definitional normalization, and the remaining obligations include the indexed E2 logarithmic-volume comparison, ambient and integral structures, theta-set and hull compatibility, and measure/weight conventions (Section 3, notice A1). The valuation-level normalization identification does not by itself carry those obligations. **Verdict: partially delivers (declared convention; derivation unserved). Q high / K high / S see notices.**

2.7 F-R8 — the second mode: visible and declared

Minimal predicate: the second mode used by the bridge is visible and its role declared (not: a complete inventory of all auxiliary modes). The second mode is the global Frobenius/log shift, the pair $(y_0, y'_0 = \varphi(y_0))$. It is named; its effect is directionally characterized (absolute values of Tate parameters rise under φ ; ATS IV, § 1.4(2), extract ll. 567–569); its asymmetric use is declared (“the lower bound will be computed using the arithmeticoid $y'_0 = \varphi(y_0)$ while the upper bound will be computed using y_0 (the upper bound is not independent of this choice)”; § 1.5, extract ll. 733–743); it is geometrically interpreted (slope comparison of Frobenius-shifted bundles; Rem. 6.10.3, extract ll. 3213–3235); and it is flagged as a correction of the paper’s own earlier release (“[The previous release of this paper had omitted this Frobenius-shifting.]”; extract l. 755). **Verdict: delivers (visibility predicate). Q high / K high / S not checked.**

Mode inventory (visibility status only; completeness is *not* claimed): the Frobenius/log shift (visible, used); the countably infinite choice of the base point y_0 (visible, unbooked; ATS III, Rem. 4.4.2, extract ll. 1723–1731); the intended-but-unavailable average over arithmeticoids (visible, unavailable; ATS IV, § 4.5(4)/(5): “one is far from being able to do this”, extract ll. 2100–2121); the auxiliary per-place quantities z_p, z'_p constrained only through their sums (visible, unbooked; ATS IV eqs. (1.5.4)/(1.5.5), extract ll. 798–819).

3 External Status Notices and Their Corrected Scope

The historical audit was run on 12 August 2026; notices A1–A3 were recorded on 13/15 August. Their former terminal interpretations are corrected here as of 2 October 2026. These notices are source-reported follow-up assessments, not proofs established by this form audit. Parts 5 and 6 are now published

working papers [11, 12]; public availability is not a mathematical acceptance or a promotion of their claims.

Notice	Corrected scope	Effect on this audit
A1 (13 Aug)	valuation-level normalization; indexed hull/measure/weight and E2 compatibility remain open	F-R2b and F-R6a remain partial
A2 (13 Aug)	ATS III tensor-box inclusion and applicable hypotheses remain open	historical proof-address finding retained; no C7 substance certification
A3 (15 Aug)	conditional effective-weight classification under fixed field/place and identity hypotheses	no unconditional F-R6a or global branch closure

Table 3: Corrected reading of the historical notices, 2 October 2026. Dates identify their origin; the current text limits their import.

A1 concerns the middle equality in ATS IV, Theorem 6.10.1 (PDF pp. 66–67), with the scope limits of § 1.5 and Remarks 6.10.2–3. The normalization supplement’s corrective v1.2 [13] distinguishes a valuation-level identification from the further choices needed to compare indexed E2 logarithmic volumes. Declared normalization does not settle ambient/integral structure, theta-set, hull, measure, weight, or ordering compatibility. A suggested Frobenius mechanism for set stability remains source-reported; it is not an established E2 transport theorem. F-NEU-1 remains open.

A2 concerns ATS III, Lemma 9.10.7.1 and Theorem 9.11.1 (PDF pp. 125 and 127). Its box-inclusion obligation is separate from A1’s ATS IV normalization/measure obligation. Correcting our generic volume paraphrase does not discharge A2 and does not refute the source-specific lemma. The currently cited Part 5 assessment [11] is retained as a dated report, not independently certified in this paper.

A3 is limited to quantities demonstrated to factor through effective product-formula weights. The reported classification requires a fixed number field and matched full set of places, real effective weights, and an identity for all nonzero elements under the stated absolute-value conventions; source-specific pointwise normalization identities impose further conditions. The full product formula alone leaves a common scalar; a pointwise normalization identity is a stronger hypothesis. Neither conclusion automatically applies to logarithmic volumes, measures, theta-sets, or hulls. The all-place hypothesis cannot silently be substituted for a non-archimedean-only place range. Parts 5 and 6 [11, 12] require their own source-bound review; no unconditional cross-container verdict, terminal closure of reading branch (c), or positive E2 theorem is imported here.

4 Synthesis (Conservative)

Table 4 follows the corrected notices and distinguishes historical address findings from current component obligations (Section 1.4).

Finding	Observed form	Obligation not certified
F-R2a / C4	explicit j^2 transport law	C6 dependency; typed evaluation identity
F-R2b / C6	declared convention; I	derived cross-container compatibility (A1)
F-R4a / C5	load and site addressed (historical predicate)	complete quantified budget
F-R4b / C5	qualitative location; partial	amount, split, residual and sign accounting
F-R6b / C7	theorem and proof addresses (historical predicate)	applicable hypothesis anchors; actual box inclusion (A2)
F-R6a / C6	evaluation available; global scope stated; partial	typed evaluation identity; indexed E2 hull/measure/weight compatibility (A1)
F-R8 / E8	Frobenius mode visible (historical predicate)	complete coverage/exclusion of order-relevant modes

Table 4: Conservative selected-rows synthesis; no current full-certificate or proof-quality score. Q/K source/criterion confidence is distinct from unchecked or open mathematical substance.

The source supplies identifiable scaling, compensation-site, and mode statements. F-R2b and F-R6a remain partial; F-R4b remains a located but unquantified budget. Historical F-R4a, F-R6b and F-R8 findings concern narrower predicates than current C5, C7 and E8. These observations permit inspection of the named obligations, not a retrospective full-contract pass. The translated-ball counterexample corrects this paper’s own paraphrase only. The A1 and A2 obligations remain distinct; the conditional A3 classification does not close either.

Level	Current scope	Remaining burden
Normalization	valuation-level identification/convention	typed/indexed comparison not established
Measure / E2	no general positive-measure classification imported	ambient, hull, measure, weight and ordering compatibility
Set stability	candidate Frobenius mechanism source-reported	no established indexed E2 set/hull transport
Box / A2	specific tensor-box hypotheses and inclusion	actual construction and applicable hypothesis anchors
Effective weights / A3	fixed-field/all-place conditional classification	applicability, place-range reconciliation; no measure consequence

Table 5: Current burden map, 2 October 2026. The former unconditional and terminal imports are withdrawn; open obligations are not new refutations of ATS.

5 Claim Boundaries

This audit makes no claim about the truth of the *abc* conjecture, the correctness of IUT, the correctness of the ATS series, or the ultimate repairability of the transport step. It judges publicly posted preprint text against a pre-specified contract form, with verbatim anchors; an unserved row predicate localizes a burden, it does not refute a theorem. The audit is restricted to the four commissioned rows; no completeness of the mode inventory, no full-certificate pass, and no aggregation of the row verdicts into a proof-quality score is claimed. The substance-level analysis and repair classifications are the subjects of the published working papers Parts 5 and 6 [11, 12]; this paper imports only explicitly limited, dated source reports, not a fresh acceptance of their mathematics.

AI Disclosure

AI systems were used for research discussion, literature and source work, mathematical analysis, text, translation, and typesetting. Earlier AI-assisted assessments remain subject to the source and claim limits stated in this working paper. For this corrective version, quality assurance includes separate source-bound mathematical and bilingual delta reviews, checks of the source versions and bibliographic identities, compilation, and visual inspection of both PDFs. These checks concern the specified corrective manuscript delta; they do not certify the ATS proof chain. No new computational experiment or whole-series proof verification is reported.

References

- [1] K. Joshi, *Construction of Arithmetic Teichmüller Spaces I*, arXiv:2106.11452v4 [math.AG], 2021 (revised 24 February 2025).
- [2] K. Joshi, *Construction of Arithmetic Teichmüller Spaces II: Proof of a Local Prototype of Mochizuki’s Corollary 3.12*, arXiv:2303.01662v3 [math.AG], 2023 (revised 24 February 2025).
- [3] K. Joshi, *Construction of Arithmetic Teichmüller Spaces II $_{\frac{1}{2}}$: Deformations of Number Fields*, arXiv:2305.10398v12 [math.AG], 2023 (revised 24 February 2025).
- [4] K. Joshi, *Construction of Arithmetic Teichmüller Spaces III: A “Rosetta Stone” and a Proof of Mochizuki’s Corollary 3.12*, arXiv:2401.13508v4 [math.AG], 2024 (revised 24 February 2025).
- [5] K. Joshi, *Construction of Arithmetic Teichmüller Spaces IV: Proof of the *abc*-Conjecture*, arXiv:2403.10430v2 [math.AG], 2024 (revised 24 February 2025).

- [6] S. Mochizuki, *Inter-universal Teichmüller Theory I: Construction of Hodge Theaters*, Publ. Res. Inst. Math. Sci. **57** (2021), no. 1/2, 3–207; *II: Hodge–Arakelov-Theoretic Evaluation*, *ibid.*, 209–401; *III: Canonical Splittings of the Log-Theta-Lattice*, *ibid.*, 403–626; *IV: Log-Volume Computations and Set-Theoretic Foundations*, *ibid.*, 627–723.
- [7] P. Scholze and J. Stix, *Why abc is still a conjecture*, unpublished manuscript, version of 16 July 2018. Available at <https://www.math.uni-bonn.de/people/scholze/WhyABCisStillaConjecture.pdf>
- [8] L. Geiger, *The Comparison in IUTchIII, Corollary 3.12: From Attempted Repair to Structural Diagnosis*, Zenodo, 2026, version v6.4. DOI: 10.5281/zenodo.23081604.
- [9] L. Geiger, *The Bridge-Contract Criterion: A Form Audit for Transport Arguments, with an Instantiation in the abc Literature*, Zenodo, 2026, version 0.2. DOI: 10.5281/zenodo.21960477.
- [10] L. Geiger, *The Bridge Certificate: A Line-Level Audit Instrument for Inter-Theoretic Transfer Arguments, with Four Applications to the abc Literature*, Zenodo, 2026, version 0.3. DOI: 10.5281/zenodo.23090371.
- [11] L. Geiger, *The Joshi ATS Series under the Bridge-Certificate Standard: A Targeted Substance Audit and an All-Place Rigidity Classification of Effective Normalization Weights*, Zenodo, 2026, version 0.2. DOI: 10.5281/zenodo.21956399.
- [12] L. Geiger, *What the Product Formula Allows: A Rigidity Theorem and an Effective-Weight Test for the ATS Height Subchain*, Zenodo, 2026, version 0.2. DOI: 10.5281/zenodo.21952487.
- [13] L. Geiger, *Global Normalization in Joshi’s Arithmetic Teichmüller Theory: What the Identification Carries — and What Carries the Theorem (DRAFT)*, Zenodo, 2026, version 1.2. DOI: 10.5281/zenodo.23092166.
- [14] L. Geiger, *The abc Landscape: Obstructions, Failed Routes, and HCT Diagnostics (DRAFT)*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21916900 (cited version 1.1, 13 August 2026, DOI: 10.5281/zenodo.21924338).